# Emre Cecanpunar

Graphics developer hacking on GPU compilers and drivers

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## Open Source Contributions

**Mesa** 2025 – Present

Independent upstream contributor to Jay

- Drive Gfx12 support in Jay as part of its expansion to older Intel GPUs.
- Contribute to Intel's BRW compiler and ANV driver internals.
- Profile compiler and driver internals to identify performance bottlenecks.
- Test changes with Vulkan CTS, Fossilize, targeted reproducers, IR inspection, and sanitizers.
- Participate in upstream design discussions, reviews, and testing.

**Linux kernel** 2025 – Present

DRM and platform drivers

- Upstreamed a five-patch hp-wmi series for x86 platform support.
- Investigate Linux DRM/GPU internals.

**Apache NuttX** 2026

Arm64 SoC bring-up and upstream fixes

- Upstreamed initial RK3588S SoC and Orange Pi 5 Pro support. Project announcement.
- Fixed display and EDID handling issues.

**Other upstream work** 2026

- Merged a heap-buffer-overflow fix in NVIDIA egl-wayland2 and merged a correctness fix in LLVM.

## Experience

**Embedded Software Engineer Intern, Baykar Technologies** Feb 2025 – Jun 2025

T3 Gemstone platform

- Initiated the Apache NuttX port to the TI AM67A SoC.
- Brought up the Cortex-R5F core, including early boot and MPU/cache setup.
- Wrote linker scripts and GPIO/UART drivers; debugged ELF loading through Linux remoteproc.

**Avionics Software Developer, Atmaca Rocket Team** Sep 2022 – May 2023

University rocketry team · TÜBİTAK competition

- Developed avionics firmware and parachute-deployment logic using Kalman-filtered sensor estimates.
- The avionics technical report received a full score from TÜBİTAK.

## Education

**B.Sc. Computer Science & Engineering, Akdeniz University** Sep 2022 – Jun 2026